

45TH ANNUAL GRAHAM GODDARD IN-HOUSE CONFERENCE/HONOURS POSTER CONFERENCE

Talks will be held in LSC P5260

Posters, lunches and breaks are in the Undergraduate Study Centre, LSC P4332.

#	Presenting Author	Title
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Tuesday, April 9th

8:50 – 10:00 AM Session 1, Chair: Ray Klein

	Tara Perrot	Introduction and brief history of convention
T1	Raymond M. Klein	On the potential for wearable technology in the modeling of vigilance and fatigue for aeronautic applications
T2	Megan Vincent	Prognostic biomarkers of seizure recurrence after first seizure and in New Onset Epilepsy
T3	Brett B. T. Feltmate	Estimating the Probability and Fidelity of Encoding during the Attentional Blink
T4	John Barresi	On Building a Person: Recent Advances and Limitations in the Construction of Robots that Attempt to Mimic Human Behavior

10:00 – 10:30 AM Coffee Break

10:30 – 11:30 AM Session 2, Chair: Sherry Stewart

T5	Matt Orr	A Systematic Review Examining In-Class Interventions for Students with Attention-Deficit/Hyperactivity Disorder in the Mainstream Classroom
T6	Amanda E.F. Hagen	Dose-Response Relationship of Alcohol on Self-Relevant Negative Thought Content in Major Depressive Disorder
T7	Anastasija Jemcov	Feasibility testing of the ABCs of SLEEPING tool as an intervention
T8	Nicole MacKenzie	Effective solutions for managing procedural pain related to vaccinations

11:30 – 1:30 PM Lunch Break and Honours Posters (Undergraduate Study Centre 4th)

Neuroscience Honours Posters

HP1	Grace Adegunju (S. Barrett)	Pharmacological and Psychological Effects of CBD on the Stress Response in Humans
HP2	Mary Archibald (S. Phillips)	Simulation-Based Education to Reduce Delays in the Treatment of Acute Ischemic Stroke

HP3	Vibhu Bhargava (A. Bombay)	Ethno-Racial Discrimination, Mental Health Stigma, and Psychological Distress Among Euro-Canadian, Indigenous, and Black Populations
HP4	Candela Bosco Saurez (J. McDougall)	Proteinase-Activated Receptor 2 Expression in the Lysophosphatidic Acid Model of Joint Neuropathic Pain
HP5	Lindsay Colyn (S. Gadbois)	Clinical Canines: Methodological Considerations in Biomedical Scent Detection
HP6	Mackenzie Cook (S. Christie)	Does Melatonin Save Neurons After Spinal Cord Injury?
HP7	Klara Doelle (A. Friedman)	Acute Convulsive Seizures, Blood-Brain Barrier Disruption and Cognitive Impairment in a Rat Model of Moderate Traumatic Brain Injury
HP8	Thalia Garvock-de-Montbrun (R. Brown)	Behavioural Analysis of the MDGA2+/- Mouse Model of Autism Spectrum Disorders at 15 Months of Age
HP9	Marwan Ibrahim (T. Akay)	Investigating the Effect of Peroneal Nerve Stimulation on the Transmission of Sensory Afferent to the Contralateral Side
HP10	David Kerr (I. Weaver)	Effects of Atrx Haploinsufficiency and Perinatal Experience on Hippocampal Development and Cell Survival
HP11	JungWoo Lee (S. Boe)	Duration of Motor Imagery Performance Impacts Cortical Excitability
HP12	Colin Mackay (Y. Zhang)	Investigating How Glutamatergic V3 Interneurons Differentially Innervate Distinct Motor Pools in the Mouse Spinal Cord".
HP13	Nicole Michaud (N. Crowder)	Optogenetic Activation of Vasoactive Intestinal Peptide Expressing Interneurons Alters Pyramidal Cell Contrast Response Functions in Mouse Primary Visual Cortex
HP14	Alysha Pelley (T. Franklin)	The Effect of Social Rank on HDAC2 Expression in the Hippocampus of Female Mice
HP15	Nicole Pickett (T. Franklin)	Abnormal Social Communication in Infant Igsf21 Mutant Mice
HP16	Lindsey Power (T. Bardouille)	Efficacy of Wireless EEG-Based Neurofeedback to Modulate Brain Activity During Motor Imagery
HP17	Taylor Saravanamuttoo (S. Johnson)	The Effect of Indirect Nature Exposure on Youth's Creative Performance
HP18	Rylan Smith (T. Perrot)	Effects of Probiotics on Anxiety-Like Behaviours, Serotonin Levels, and BDNF Levels
HP19	Marjorie Sollows (L. Phillmore)	The Effects of Breeding Condition on DNA Methylation in European Starlings
HP20	Paula Torres Muñoz (R. Brown)	Neurobehavioural Characterization of the Nrnx1 +/- Mouse Model of Autism Spectrum Disorder
HP21	Jennika Veinot (A. Newman)	The N170 and Skilled Reading in English and French Immersion Students: An ERP Study
HP22	Hailey Zwicker (S. Boe)	Examining the Effect of High Intensity Interval Training (HIIT) on Corticospinal Excitability and Consolidation of Motor Learning
HP23	Mikaela Friedrich (K. Semba and S. Deurveilher)	Analysis of circadian locomotor rhythms in the stable tubule only polypeptide (STOP) null mouse model of schizophrenia

Psychology Honours Posters

HP24	Seyi Ade Ojo (T. Juckes)	University-Aged Students' Beliefs About Rape Myths for Self, Same-Gender and Opposite-Gender Peers
HP24	Armita Amirimoghadam (R. Brown & J. Wang)	Effect of Noise-Induced Hearing Loss on Spatial Learning and Memory of C57BL6/J Mice
HP26	Jocelyn Brown (D. Gilin-Oore)	The Impact of Social Anxiety on Perceived Conflict Handling Style in Negotiations
HP27	Olivia Capozzi-Davis (C. Moore)	Examining Children's Judgements of Care and Sanctity Transgressions
HP28	Emily Cote (H. Deacon)	In Between the Letters: How Do We Spell? Assessing the Effect of Morphological Complexity on Handwriting Speed
HP29	Kyra Farrelly (S. Stewart)	Do Coping Motives Mediate the Relationship Between PTSD Symptoms and Cannabis Craving After a Personal Trauma Cue?
HP30	Anastasia Gvozdeva (C. Moore)	Sensitivity to Moral Foundations of Two- and Four-Years-Old Children
HP31	Agnes Hardardottir (S. Stewart)	The Effect of Cigarette Packaging on Attention to Graphic Warnings: A Controlled Study
HP32	Graham Hood (Tim Juckes & S. MacKinnon)	Heterosexual University-Aged Students' Perceptions of Rape and Consent in Male-on-Female Sexual Assault Situations
HP33	Brett Hopkins (S. Stewart)	Smoking Behaviour in a Controlled Setting: The Roles of Plain Packaging and Illness Sensitivity
HP34	Abigail Howard-Gosse (C. Moore)	Getting Through Those Piles of Readings: How Students With a History of Reading Difficulties Manage Reading in University
HP35	Olivia Innes (C. Moore)	Comparing Children's Sensitivities to the Moral Foundations of Authority and Loyalty
HP36	Kiyana Kamali (T. Perrot)	The Effects of Probiotics on Maternal Care Behavior Quality and Consistency, and Subsequent Offspring Anxiety-Like Behaviours
HP37	Manasi Karkada (I. Smith)	Heterogeneity, Hyperlexia, and How Children With Autism Spectrum Disorder Read
HP38	Adith Mudduraju (G. Eskes)	If You Mind, Does It Matter?"
HP39	Michaela Purdon (R. Brown)	Behavioural Characterization of the MDGA2+/- Mouse Model of Autism Spectrum Disorder at Nine Months of Age
HP40	Connor Reid (S. Jacques)	The Relation Between Receptive Vocabulary and Object Permanence in Dogs Canis familiaris

1:30 – 2:30 PM Session 3, Chair: Nathan Crowder

T9	Elise Aronitz	Examination of NMDA receptor subunit composition in the developing cat visual cortex
T10	Jared T. Shapiro	Optogenetic Modulation of the Excitatory-Inhibitory Balance of Mouse V1 Responses to Drifting Sinusoidal Gratings

T11	Steve Shaw	Conduction velocity and synaptic delay at photoreceptor $\rightleftharpoons$ amacrine reciprocal synapses in the fly's eye
T12	Jasmine Aziz	Role of working memory in visual search among older adults

2:30 – 3:00 PM Coffee Break

3:00 – 4:30 PM Session 4, Science Slam, Chair: Tara Perrot

SS1	Kyra Farrelly	Alcohol and Cannabis Co-Use
SS2	Isaac Romkey	Jazz Training and Working Memory
SS3	Samantha Good	Implementing Binocular Rivalry to Study Aphantasia
SS4	Gracious Kasheke	Genetics: The Secret Recipe to Understanding Psychiatric Disorders
SS5	Heather Ouellette	Green Space Effects on Mental Health
SS6	Jenna Andrews	The Effects of Keeping A Secret
SS7	Griffin Mumby	The Benefits of Exercise in Alzheimer's Disease
SS8	Agnes Hardardottir	The Effect of Bullying on the Adolescent Brain
SS9	Mallory Whebby	Rhythmic Auditory Cueing and Parkinson's Disease
SS10	Jeremy Bartholomeusz	Venom – The Future of Cancer Treatment Administration
SS11	Qendresa Sahiti	Concussion in Soccer

4:30 PM onwards – University Club Pub

Wednesday, April 10th

9:00 – 10:00 AM Session 5, Chair: Ian Weaver

T13	Emre Fertan	Understanding the role of amyloid beta in Alzheimer's disease pathogenesis: teaching an old dog new tricks
T14	Ryan V. Wheeler	Effects of LPS-Induced Maternal Immune Activation on Expression of Autism-Linked Genes and Autism-Like Behaviours in Neonates and Adulthood
T15	Gloria Rodrigues	Effects of forebrain ATRX deficiency in C57/B6 male mice on offspring ATRX expression and forebrain development: maternal nurture meets offspring nature
T16	Filip Kosel	Phenotypic contagion of aggression in male 5xFAD mice

10:00 – 10:30 AM Coffee Break

10:30 – 11:45 AM**Session 6, Chair: Helene Deacon**

T17	Colin McCormick	Reducing Cognitive Demand to Explore the Speed Accuracy Trade-off in Endogenous and Exogenous Temporal Cueing
T18	Anjali Pandey	What Information Escapes an Instruction to Forget?
T19	Sinan Aslam	On the relationship between spatial compatibility effects and Inhibition of Return
T20	Kiera O'Neil	Auditory Language Processing in the Visual Cortex of Blind Individuals
T21	Lindsay Rosenberg	How Does Morphological Awareness Contribute to Reading Comprehension in Beginning Readers: A Mediation Analysis

11:45 – 1:15 PM Lunch Break and In-House Posters

P1	Tasnim Ali	The Relation Between Hot and Cool Inhibitory Control and Temperament in 3- to 7-Year-Olds
P2	Ilne Barnard	An examination of spectrin protein across development in human visual cortex, and following monocular deprivation in cat lateral geniculate nucleus.
P3	Spencer L. Brown	Behavioural characterization of the Nr1h1 +/- mouse model of autism spectrum disorder
P4	Sara Cochrane	Why Don't We Spend Time In Nature?: A Thematic Theory-Driven Analysis
P5	Brianne Derby	The Emergence of Envy and Schadenfreude Through Social Comparison and the Disappointing Gift Paradigm
P6	Richard S. Drake	Seeing the forest for the trees in large cognitive datasets
P7	Evan F. Forward	Relating Pupillometric and Behavioural Indices of Effort in Item-Method Directed Forgetting
P8	Samantha Good	Searching for a Target: Do Older Adults Check the Template More Often?
P9	Linda Hutchinson, Gabrielle Moulton & Oliver Schnare	Language and Cognition in Dogs (<i>Canis Familiaris</i>)
P10	Aydan Idrisova	The Influence of Varying Foreperiods on Error Rate and Reaction Time
P11	Kirsten M. Lebel	Even Further Evidence in Favor of Prior Entry from Endogenous Attention to a Location in Space: A Replication Study
P12	Kate MacDiarmid	On the independence of attentional mechanisms in temporal orienting: a replication study
P13	Alexandra M. Ryken	Reading comprehension, prosodic sensitivity, and the role of prosodic cues in text

P14	Catherine Reeve	Knowledge of Spoken Words by Domestic Dogs: A New Instrument for Use With Dog Owners
P15	Kaelen Watters	Forgetting on purpose; How and what do we intentionally forget?
P16	Francesco Usai	Investigating the neural bases of self-monitoring of speech in people with aphasia: an ERP study

1:15 – 2:15 PM Special Session 7, Facilitator: Aaron Newman

THE FUTURE OF PSYCHOLOGY AND NEUROSCIENCE AND DALHOUSIE

Our department is at an exciting point in its history, with recent changes in leadership of our undergraduate and graduate programs, a new Chair, proposed changes to our graduate program, and an increasing emphasis on communications and outreach. Members of the department from 1st year undergraduate to emeritus faculty are invited to participate in a town hall-style discussion on the future of our field, the needs of the trainees we serve, and how we can be most relevant and impactful in the coming decade.

2:15 – 2:45 PM Coffee Break

2:45 – 4:00 PM Session 8, Chair: Aaron Newman

T22	Sean Mackinnon	The Software Wars: SPSS vs. R (and beyond)
T23	Lyam M. Bailey	Sinister Participants: Is modern neuroimaging a right-handed tool?
T24	Swasti Arora	Does Question Phrasing Matter? A mixed-methods analysis of Student Ratings of Instruction
T25	Colin Conrad	The Psychophysiology of Mind Wandering During Online Lectures
T26	Helene Deacon	How do children become good word readers: The role of learning about how print works.

4:00 PM Closing Comments

In-House Abstracts

TALK ABSTRACTS

TITLE: On the potential for wearable technology in the modeling of vigilance and fatigue for aeronautic applications

AUTHORS: Raymond M. Klein, Brett Feltmate, Austin Hurst, Maelle Kopf & Jean-François Gagnon

ABSTRACT: Fatigue-related decreases in vigilance (hypovigilance) and their negative effects on performance are widely recognized as posing a serious risk to aviation safety. Scientists at Thales Research & Technology (TRT) Canada (Quebec) collaborated with the Klein laboratory to design and conduct a research project in which participants performed tasks over an 8 hour period during which a variety of physiological, oculomotor and performance parameters, and self-reports of fatigue, were collected either continuously or regularly. Thales scientists then developed a mathematical model that from the physiological data could predict fatigue-related hypovigilance; a model that was successful when generalized to “new” data and that was recently tested in a more ecologically valid scenario (an intercontinental flight). In a followup project the Dalhousie-Thales team plans to test the efficacy of mitigation measures of hypovigilance based on this model’s on-line analysis of wearable data collecting during performance on a driving-like task.

TITLE: Prognostic biomarkers of seizure recurrence after first seizure and in New Onset Epilepsy

AUTHORS: Megan Vincent (Supervised by Dr. Omisade and Dr. Fisk)

ABSTRACT: This study seeks to establish prognostic markers of developing epilepsy after a first seizure (FS) or seizure recurrence despite treatment in patients with new onset epilepsy (NOE). Predicting seizure recurrence may facilitate early treatment and reduce disease burden. Our primary objective is to determine if cognitive impairment, mood, and socioeconomic status (SES) can predict seizure recurrence after FS or NOE. We will select 50 patients with NOE or FS at the Halifax First Seizure Clinic. Baseline testing will be conducted and includes measures of major cognitive areas and mood symptoms. Follow-up testing using the same measures will be conducted after 1 year. Chart review will determine 1) seizure recurrence since initial visit 2) baseline results and 3) clinical and demographic measures (diagnostic test results, seizure frequency, SES). The patients will be divided into two groups based on seizure recurrence since baseline: no seizures vs 1+ seizures. Regression analyses will determine if baseline measures are predictive of seizure recurrence.

TITLE: Estimating the Probability and Fidelity of Encoding during the Attentional Blink

AUTHORS: Brett B. T. Feltmate, Ralph S. Redden, Raymond M. Klein

ABSTRACT: When participants are required to report the presence or identity of two targets presented in close temporal sequence, they are often able to recall that of the first (T1), but not the second (T2). Referred to as the ‘attentional blink’, this phenomenon is thought to reflect the tying up of resources needed for target processing by T1 leaving them unavailable for T2 processing. Herein we argue that in conventional ‘blink’ tasks success is measured in dichotomous fashion, thereby conflating two potential sources of recall error: the probability of, and the fidelity to which, T2 is encoded into visual short-term memory. We attempted to generate independent measures of probability and fidelity by making colour our relevant target domain. Target colouring was randomly sampled from within a colour wheel, and responses were made by clicking within the same wheel allowing us to record the degree of error in a given response. Obtained estimates of our parameters are assessed in exploratory fashion and their implications are discussed herein. All pre-registered materials can be found on the OSF at <https://bit.ly/2BSHIKH>.

TITLE: On Building a Person: Recent Advances and Limitations in the Construction of Robots that Attempt to Mimic Capacities of Human Persons

AUTHORS: John Barresi

ABSTRACT: Based on criteria of personhood that robotic agents should pass for them to match us in social intelligence, I describe some recent successes as well as difficulties for the future with respect to these criteria. I give illustrations of research in several areas where advances have been made: 1) Evolutionary robotics; 2) Developmental robotics; 3) Knowledge sharing; 4) Robo-science.

TITLE: A Systematic Review Examining In-Class Interventions for Students with Attention-Deficit/Hyperactivity Disorder in the Mainstream Classroom

AUTHORS: Matt Orr, Sarah Brine, Nicole Ali, and Penny Corkum

ABSTRACT: It is difficult for teachers to meet the needs of students with ADHD and in-class interventions could help to provide effective support. However, access to information about evidence-based in-class interventions is limited. A systematic review was conducted to determine what evidence-based in-class interventions for ADHD are present in the literature. Fourteen studies examining in-class interventions for ADHD were identified and reviewed. These studies featured a variety of interventions such as tactile cueing, self-management strategies, and peer monitoring. Studies were of moderate-to-high methodological quality. The available evidence does not meet the criteria of evidence-based interventions, suggesting that the summarized studies cannot be confidently recommended as in-class interventions for students with ADHD.

TITLE: Dose-Response Relationship of Alcohol on Self-Relevant Negative Thought Content in Major Depressive Disorder

AUTHORS: Amanda E.F. Hagen, Sherry H. Stewart

ABSTRACT: Major depressive disorder (MDD) and alcohol use disorder co-occur and are associated with poorer outcomes. Of possible underlying mechanisms, the self-medication hypothesis posits that individuals with low mood may drink to improve their mood. There is evidence that low doses improve mood, but that mood decreases at higher doses. Individuals with MDD may find alcohol reinforcing due to the alleviation of cognitive symptoms, e.g. negative self-thinking. One study showed that alcohol dampens negative self-relevant thinking at moderate doses in individuals with mild depression symptomology. Extending this, the current study will study those with MDD and examine different doses of alcohol. We hypothesize the relationship between mood and dose will be extended to cognitive factors. Individuals with depression will be randomly assigned to either no alcohol, low alcohol, or high alcohol conditions, before completing the word task. This study will help understand mechanisms of this comorbidity to inform specific interventions.

TITLE: Feasibility testing of the ABCs of SLEEPING tool as an intervention

AUTHORS: Anastasija Jemcov, Dr. Penny Corkum

ABSTRACT: Behaviourally-based sleep problems are common among school-aged children, are highly treatable, but due to a number of barriers, very few children receive evidence-based behavioural interventions. The ABCs of SLEEPING tool was developed to facilitate the assessment of sleep problems by identifying healthy sleep practice strengths and areas in need of improvement, and to provide parents with evidence-based recommendations. Using Bowen and colleagues (2010) “Does It Work?” feasibility framework, the tool will be tested for its acceptability, implementation, and limited-efficacy, using a cohort, pre-test/post-test design.

TITLE: Effective solutions for managing procedural pain related to vaccinations

AUTHORS: Nicole MacKenzie (Supervisor: Dr. Christine Chambers)

ABSTRACT: Children often experience pain during vaccinations, however many parents are unaware of evidence-based strategies available for managing vaccination pain. A knowledge translation (KT) resource was previously developed to summarize evidence-based strategies for managing children’s vaccination pain. This study aimed to examine parents’ perceptions and plans to use this KT resource using the Information Assessment Method (IAM) to assess the relevance, cognitions, utility, and benefits of the resource. Attitudes toward vaccines were also measured. 312 parents completed this survey and 97% demonstrated little to no vaccine hesitancy. Vaccine attitudes were negatively correlated with relevance of the KT resource ($r_s = -.17, p = .003$) and intention to use the information ($r_s = -.14, p = .012$), indicating that vaccine accepting parents were more likely to be receptive and planned to use the strategies. This study highlights key factors which impact parents’ plans to use KT resources for vaccination pain management.

TITLE: Examination of NMDA receptor subunit composition in the developing cat visual cortex.

AUTHORS: Elise Aronitz & Kevin R. Duffy

ABSTRACT: Many brain regions exhibit a period during early development called a ‘critical period’, when the capacity for neural plasticity is high, and neural circuitry is rapidly modified in response to neural activity. In the primary visual cortex of mammals, this capacity for rapid modification has allowed for the refinement of functional characteristics of neural circuitry that optimize visual performance. Following waning and closure of the critical period, neural circuitry within the visual system exhibits minimal capacity for modification. Changes in plasticity that occur throughout development are thought to occur as a result of a change in the subunit composition of the NMDA receptor. We will employ a novel approach to quantifying the ratio of NMDA receptor subunits across development that involves multiplex fluorescence imaging.

TITLE: Optogenetic Modulation of the Excitatory-Inhibitory Balance of Mouse V1 Responses to Drifting Sinusoidal Gratings

AUTHORS: Jared T. Shapiro, Jillian L. King, Nathan A. Crowder

ABSTRACT: The primary visual cortex (V1) has served as a model brain region for studying cortical information processing because it is the first cortical stage of the visual pathway, and a region where several novel computations arise. Recent work has investigated the neuronal circuits underlying receptive field properties of mouse V1 neurons due to the many genetic tools available in this species that enable cell-specific targeting and perturbation. One such genetic tool is optogenetics, which has been previously used to demonstrate how distinct classes of inhibitory GABAergic interneurons can affect the excitatory drive of V1 pyramidal cells. Importantly, this past work has focused on optogenetic modulation of the onset and sustained responses of V1 pyramidal cells, but has given no attention to the pattern of activity evoked after photostimulation is terminated. We used optogenetics and *in vivo* electrophysiology to investigate the role of parvalbumin-expressing interneurons (PV+) in V1 circuitry, and specifically how these interneurons affect pyramidal cell after-responses. We have demonstrated that termination of photostimulation produces a “rebound effect” on pyramidal cell activity, which facilitates their after-responses to flashed bars and sinusoidal gratings. We investigate whether this “rebound effect” is also produced from drifting sinusoidal gratings.

TITLE: Conduction velocity and synaptic delay at photoreceptor $\rightleftharpoons$ amacrine reciprocal synapses in the fly’s eye.

AUTHORS: Steve Shaw

ABSTRACT: The network of amacrine neurons in the first synaptic zone (lamina) of the fly’s compound eye makes reciprocal feedback synapses into each visual column, upon the six photoreceptor terminals (R1-R6) that drive the amacrine in the first place, Am $\rightleftharpoons$ R1-6. Previous results indicate that following light activation, the amacrine net activates 1-2 inhibitory impulses,

which rapidly but transiently suppress R1-6 terminals. This causes R1-6 input to the main lamina output cells L1-3 to become highly phasic. Fast active presynaptic suppression of R1-6 can be detected starting at moderate light intensity in the rapid downstroke of the Early Positive Response (EPR), the field potential from L1-3. The delay between light input at one side of the eye and complete response suppression at the other side is ~2 milliseconds, comprising the impulse conduction delay followed by the monosynaptic delay: these two values are not known separately. By recording feedback oscillations caused by the Am $\rightleftharpoons$ R1-6 reciprocal interaction at opposite ends of the eye, the conduction velocity along the amacrine network is estimated at ~1.3 meters per second. Maximum horizontal conduction delay at equatorial level is then ~0.6 milliseconds. Average synaptic delay becomes 1.4 milliseconds, but may be different for R1-6 and Am.

TITLE: Role of working memory in visual search among older adults

AUTHORS: Jasmine Aziz, Raymond Klein, Gail Eskes

ABSTRACT: Visual search and working memory abilities decline with age, which can interfere with daily activities. During visual search, working memory helps us remember target features and previously searched locations. Thus, we investigated whether working memory abilities would predict visual search efficiency in healthy older adults. Thirty participants (55-78 yrs) completed working memory measures and a visual search task. We varied working memory load by manipulating whether a reference picture of the target (i.e., target template) was displayed during the search. We also recorded eye movements to determine how frequently participants looked at the target template. Visuospatial working memory abilities predicted better conjunction search performance, and verbal working memory abilities interacted with the template presence/absence, although independently of template fixation time. Findings clarify how visual search abilities relate to working memory in older adults and could be helpful for developing ways to support these abilities in aging and relevant clinical populations.

TITLE: Understanding the role of amyloid beta in Alzheimer's disease pathogenesis: teaching an old dog new tricks

AUTHORS: Emre Fertan, Richard E. Brown, Ian C.G. Weaver

ABSTRACT: Alzheimer's disease (AD) was first described by Dr. Alzheimer in the beginning of the 20th century, based on the observation of "military fossa" made of a "special substance". In the 1980's this substance was identified as amyloid beta, which gave birth to the amyloid cascade hypothesis. However, for many years, the exact role of amyloid beta in the deficits seen in AD remained unknown and many believed that it functioned as a physical barrier in the synapse, hindering the chemical transmission. To explain the other, seemingly unrelated, symptoms, many other theories have been proposed regarding immune dysfunction, oxidative stress, tau phosphorylation. Recently, studies began to appreciate the role of amyloid beta in these "unrelated" disease mechanisms. My talk will focus on our recent work, in which we studied the connection between the increase in amyloid beta, oxidative stress, immune dysfunction and tau phosphorylation in terms of the kynurenine and TRX reduction pathways.

TITLE: Effects of LPS-Induced Maternal Immune Activation on Expression of Autism-Linked Genes and Autism-Like Behaviours in Neonates and Adulthood

AUTHORS: Wheeler RV, Weaver ICG, Franklin TB.

ABSTRACT: Epidemiological studies have revealed a link between maternal immune activation (MIA) and autism-spectrum disorder (ASD) risk. To date, the literature has primarily focused on the effects of LPS-MIA in adult male mice; there are few reports describing the behavioural effects of LPS-MIA in female mice or early in development. Further, the involvement of the mammalian target of rapamycin (mTOR) signaling pathway, a risk factor for ASD, has not been studied in this model. In the current study, we injected pregnant C57BL/6J mice with 75µg/kg of LPS on embryonic day 11.5 and 12. Male and female offspring were tested during development and at adulthood for the three core symptoms of ASD: social communication, social interaction, and repetitive behaviours. Additionally, the expression, protein abundance and activity-induced phosphorylation of mTOR and its downstream effect on post-synaptic density 95 (PSD95) was analysed in hippocampal and cortical areas of offspring at postnatal day zero (P0) and at P130.

TITLE: Effects of forebrain ATRX deficiency in C57/B6 male mice on offspring ATRX expression and forebrain development: maternal nurture meets offspring nature

AUTHORS: Gloria Rodrigues, Kristen Lee, Hillary Maillet, Donna Goguen and Ian Weaver

ABSTRACT: Early childhood experience and degree of parental-infant attachment influence the developing brain, notably in brain areas that support stress regulation, cognition and social behaviours. Siblings of children with autism have an increased relative risk of autism themselves, suggesting a role of gene x environment interactions in the etiology of autism. Although the exact mechanism is not completely understood, the autism risk gene, *Atrx*, is regulated by maternal behaviour. ATRX is a chromatin remodeling factor that is essential for brain development and plasticity; even small changes in expression and signaling can cause synaptic deficits and contribute to neurodevelopmental disorders. My talk will focus on our recent work, in which we studied the connection between maternal care and the development of brain regions relevant to autistic-like behaviours in the adult offspring of litters containing either wildtype female and male mice or wildtype female and *Atrx* hemizygous male mice. Our findings demonstrate a persistent effect on forebrain growth, *Atrx* gene expression and promoter methylation, not only in the ATRX deficient male mice, but also in their heterozygous female litter mates. These results suggest a role of early life environment and ATRX expression in pathological mechanisms underlying long-term chromatin plasticity and neural growth relevant to neurodevelopmental disorders such as autism.

TITLE: Phenotypic contagion of aggression in male 5xFAD mice

AUTHORS: Kosel, F., Hamilton, J., Godin, V., Harrison, S., & Franklin, T.B.

ABSTRACT: Alzheimer's disease (AD) is a progressive neurodegenerative disorder that is characterized by a number of cognitive and behavioural effects, including increased agitation and aggression, as well as depressive symptoms, anxiety, irritability, and other neuropsychiatric symptoms. We previously found that females of the 5xFAD transgenic mouse model of Alzheimer's disease exhibit decreased social investigation that mimics the social withdrawal observed in AD patients. This study examines whether transgenic 5xFAD males also exhibit decreased social investigation, and whether transgenic 5xFAD males model the increased aggression observed in many male AD patients. We tested male 5xFAD mice in behavioural tests for sociability, social novelty preference, and free social interaction, as well as examining home-cage interactions. Results suggest that transgenic males exhibit decreased social interest and altered aggressive behaviours compared to wild-type controls, suggesting that they may provide a useful model of aggression and social withdrawal in AD patients.

TITLE: Reducing Cognitive Demand to Explore the Speed Accuracy Trade-off in Endogenous and Exogenous Temporal Cueing

AUTHORS: Colin McCormick, Ralph Redden, Raymond Klein

ABSTRACT: Recently (McCormick, Redden, Lawrence, and Klein, 2018), we observed the independence of exogenous and endogenous temporal attention when implementing a novel signalling method (Lawrence & Klein, 2013) within the canonical temporal cueing paradigm (Kingstone, 1992). However, in an attempt to alleviate task demand, we were limited to the analysis of detection responses instead of the more informative discrimination responses. This is because we were unable to obtain TCEs in the endogenous condition when using discrimination responses, which we suspect was due to the large cognitive load generated from a combination of stimulus and task demands. For the current proposed experiment, we have modified methodological components to alleviate cognitive demand to allow for the implementation of discrimination targets in the Kingstone paradigm, providing opportunity to analyze speed-accuracy trade-off differences for endogenous and exogenous temporal attention. All project materials can be found on the Open Science Framework (<https://OSF.io>).

TITLE: What Information Escapes An Instruction to Forget?

AUTHORS: Anjali Pandey, Supervisor: Tracy Taylor-Helmick

ABSTRACT: The directed forgetting effect is a phenomena characterized by participants recalling or recognizing more remember instructed than forget instructed items. However, not all forget instructed items are removed from memory and while it can be concluded that an instruction to forget decreases the probability of a memory representation later being retrieved, standard recall or recognition tests reveal nothing about the fidelity of those representations. In looking at what aspects of words (sound, appearance,

meaning) escape initial attempts to prevent encoding when the intention is to forget, we expect that the retrieved representations of forget instructed words will be impoverished on the aspects investigated. The results of a recognition test following a typical item-method study phase will be analyzed to compare the mean false alarms of critical foils against those of unrelated foils, with a greater false alarm rate to critical foils suggesting that the single shared aspect between the study and test word is part of what was encoded.

TITLE: On the relationship between spatial compatibility effects and Inhibition of Return

AUTHORS: Sinan Aslam, Ralph S. Redden, Raymond M. Klein

ABSTRACT: Inhibition of return (IOR) is an increase in reaction time when orienting to targets that have been previously attended to. The Simon effect is a tendency to respond faster to targets presented in a location spatially compatible with the response. We explored the consequences of the Simon effect on targets suffering from IOR when the frequency of targets is altered. This was accomplished by having participants distributed amongst three conditions where the Simon target was presented in either 20%, 50% or 80% of trials. The remaining trials required a reporting the direction of a centrally-presented arrow. We expect to see a linear change the Simon effect as a consequence of target frequency, where the Simon effect may increase with more directional responses (80% arrows), or the Simon effect may increase with more peripheral events (80% Simon targets). Furthermore, the degree to which IOR impacts this change will be explored.

TITLE: Auditory Language Processing in the Visual Cortex of Blind Individuals

AUTHORS: Kiera O'Neil and Aaron Newman

ABSTRACT: In people who are blind, the visual cortex takes on a new linguistic function and becomes involved in processing speech, something that is not seen in sighted people. However, the exact role of the visual cortex in auditory language processing in people who are blind is unclear. Establishing how the visual cortex is integrated into the existing language network may shed light on its specific role. We collected fMRI data from two groups of participants, congenitally blind adults ($n = 6$) and sighted controls ($n=10$). Participants made matching decisions about auditorily presented pairs of words (meaning, rhyme, speaker) and performed a n-back task. We found visual cortex activity in the blind group but not the sighted group during linguistic and verbal memory tasks. Our next step is to determine how the visual cortex is connected to other regions in the language network (via dynamic causal modelling) and how those connections might change depending on what type of language processing is occurring.

TITLE: How Does Morphological Awareness Contribute to Reading Comprehension in Beginning Readers: A Mediation Analysis

AUTHORS: Rosenberg, Lindsay, Levesque, Kyle, & Deacon, Helene

ABSTRACT: Reading comprehension, the ability to comprehend the message that written words convey (Nation, Cocksey, Taylor, & Bishop, 2010), is the primary goal of reading. Morphological awareness is the awareness of, and ability to manipulate, meaningful units in spoken language (*teach* + *-er* = *teacher*) (Carlisle, 1995). Morphological analysis is using awareness to infer the meaning of a word (e.g. *treelet* = small tree, because *-let* means small). Both morphological awareness and analysis are associated with reading comprehension. Building on previous research with grade 3 students, this study will examine how morphological awareness contributes to reading comprehension at the grade 1 level. The study will assess whether morphological awareness and analysis are separate dimensions for beginning readers and whether morphological analysis and word reading ability are mediating factors in the relationship between morphological awareness and reading comprehension. This study will assess 200 grade 1 students on morphological analysis, morphological awareness, word reading, and reading comprehension. It is predicted that morphological analysis will have a larger mediating role compared to word reading.

TITLE: The Relation Between Hot and Cool Inhibitory Control and Temperament in 3- to 7-Year-Olds

AUTHORS: Tasnim Ali & Sophie Jacques

ABSTRACT: *Temperament* refers to children's individual differences in emotional, motor, and attentional reactivity and their ability to self-regulate those reactions (Rothbart, 2007). In this study, we examine the relation between Rothbart's three temperament dimensions and their individual relations with *hot* and *cool inhibitory control* processes in children aged 3-7 years. Hot and cool inhibitory control processes are cognitive processes that allow children to withhold dominant responses to stimuli when required to solve novel and difficult problems in affective and neutral contexts, respectively. Three- to 7-year-olds were given hot and cool versions of three inhibitory control tasks: the Flanker task, the Stroop task, and the Go-No-Go task and their parents were given the short form of Rothbart's Children Behaviour Questionnaire (Putnam & Rothbart, 2006). We predicted a negative correlation between scores on the two affective dimensions of temperament (negative affect and surgency), and performance on both versions of the inhibitory control tasks, with stronger negative correlations for the hot versions. However, we predicted a positive correlation between performance on both versions of the inhibitory control tasks, and effortful control, the self-regulatory dimension of temperament. Only some of the predicted correlations were significant with the sample we collected to date.

TITLE: The Software Wars: SPSS vs. R (and beyond)

AUTHORS: Mackinnon, S. P.

ABSTRACT: There is increasing, vociferous debate on which statistical software to use in the social sciences. On one side are the SPSS users, who are accused of using antiquated methods, having non-transparent workflows, and producing extremely ugly graphs.

On the other side are the R users, who can in theory create better final products, but in practice take an enormous amount of time unless they are experienced coders (who frequently use their remaining time to tell SPSS users why they are bad at statistics). I will discuss my experiences with both software types in research and teaching, with an eye to finding solutions that are (a) reproducible, (b) make use of modern statistical practices; and are (c) easy to learn and time-efficient. I will highlight a new free point-and-click statistical software “Jamovi” that provides a hybrid of point-and-click efficiency with the flexibility of R coding that may prove useful for teaching.

TITLE: Sinister Participants: Is modern neuroimaging a right-handed tool?

AUTHORS: Lyam M. Bailey, Laura E. McMillan, & Aaron J. Newman

ABSTRACT: Approximately 10% of humans are left-handed or ambidextrous (adextral). It has been suggested that this demographic is largely avoided by researchers within the neuroimaging community. To date, however, no formal effort has been made to quantify the extent to which adextrals are excluded from neuroimaging-based research. Here we aim to address this question in a review of over 1,000 recent articles published in high-impact, peer-reviewed, neuroimaging-focused journals. Specifically, we sought to ascertain whether, and the extent to which adextrals are underrepresented in neuroimaging study samples. Handedness data was available for over 30,000 research subjects; only around 3% of these individuals were adextral – considerably less than the 10% benchmark one would expect if neuroimaging samples were truly representative of the general population. This observation was generally consistent across different areas of research, but was modulated by the demographic characteristics of participants. The epistemological and ethical implications of these findings are discussed.

TITLE: Does Question Phrasing Matter? A mixed-methods analysis of Student Ratings of Instruction.

AUTHORS: Swasti Arora, Jennifer Stamp, Leanne Stevens, Marcus Cormier & Gisela Jones

ABSTRACT: The efficacy of Student Ratings of Instructions (SRIs) is often questioned by teaching professionals. Much research has been done on this topic, and meta-analyses have supported the use of SRIs as indicators for teaching effectiveness and tools for course development (Cohen et al. 1981). However, the colloquial discussion shows a divided opinion among instructors and faculty (Theall, M. (2001). The current study explores the importance of SRI question structure as a predictor of course feedback depth and content. Facilitated by the use of qualitative analysis software, 2,780 student responses collectively were coded from 2015 and 2016 SRIs from an Introductory Psychology course. A codebook of 23 codes, associated with 7 overarching themes and 7 contexts to which codes could correspond was established. Rates of code occurrence and co-occurrence were then calculated between open-ended questions and questions which were structured to address specific aspects of course material and development.

TITLE: The Psychophysiology of Mind Wandering During Online Lectures

AUTHORS: Colin Conrad and Aaron Newman

ABSTRACT: Mind wandering is a construct which describes presence of task unrelated thoughts, and is often identified as an experience that is detrimental to the efficacy of education. However, there has been relatively little research into the impact on mind wandering on education because it is difficult to measure. In this presentation, we will describe the results of two experiments that establish the foundation a covert auditory attention EEG measure which is correlated with reported mind wandering states. In the experiments, participants watched an online lecture video while task-irrelevant audio tones were played in the background. We found significant differences in Event Related Potentials evoked from auditory stimuli during reported mind wandering. The results present a foundation for a larger project in exploring the potential of real-time EEG detection for use in the improvement of education technologies or other information technologies, as well as new types of brain-computer interfaces.

TITLE: How do children become good word readers: The role of learning about how print works.

AUTHORS: Deacon, S. Hélène & Conrad, Nicole

ABSTRACT: A core question in models of word reading lies in which skills predict individual differences in reading acquisition: specifically, why is it that some children struggle to learn to read, while others acquire reading skill with apparent ease? Most models point to orthographic processing, or knowledge of letter- and word-patterns, as key to children's progress in word reading development. Unfortunately for these models, empirical evidence has not borne out this prediction, with multiple studies now showing that orthographic processing is an outcome, rather than a determiner of word reading, at least when measured with traditional tasks. In this talk, I present a new theoretical conceptualisation of orthographic processing, one that includes children's first steps into learning about how printed words work. I also show new evidence that children's appreciation of these early aspects of orthographic features predicts their progress in learning to read.

POSTER ABSTRACTS

TITLE: An examination of spectrin protein across development in human visual cortex, and following monocular deprivation in cat lateral geniculate nucleus.

AUTHORS: Ilne L. Barnard, Lauren J. Goodyear, Rhiannon N. Holt, Aliana Kanji, Katelyn MacNeil, Devan R. Pancura, Madison Presunka, Alexandra C. Proctor, Madelaine K. Robertson, Kevin R. Duffy

ABSTRACT: Disruption to normal vision during so-called critical or sensitive periods can modify the structure of neurons and neural connections, and can lead to a severe vision impairment called amblyopia. Decades of research on synaptic plasticity mechanisms have revealed the NMDA receptor plays a critical role in mediating the refinement modification of visual neural circuitry during the critical period. Spectrin is an actin-binding protein that has been implicated as a regulator of NMDA channel activity through its stabilizing effect on the receptor. Spectrin therefore is understood to play an enabling role in visual neural plasticity. In the current study, we quantified the level of spectrin protein in samples of human primary visual cortex (V1) across the lifespan, and we also measured the effect of monocular deprivation on spectrin labeling within the lateral geniculate nucleus (LGN). Our examination of human V1 revealed maximum spectrin at ages when plasticity levels are high. Further, we discovered that a brief period of monocular deprivation, sufficient to provoke structural modification, reduced spectrin within layers of the LGN serving the deprived eye. In aggregate, these results demonstrate a relationship between spectrin and neural modification in V1 and LGN, consistent with playing a regulatory role in visual system plasticity.

TITLE: Behavioural characterization of the *Nrxn1* +/- mouse model of autism spectrum disorder

AUTHOR: Spencer L. Brown, Qendressa Sahiti, Richard E. Brown

ABSTRACT: Autism spectrum disorder (ASD) is a neurodevelopmental disorder that is characterized by deficits in social interaction, increased repetitive and stereotypical behaviours, and abnormal responses to sensory stimulation. Previous research has linked the causation of ASD to complex gene-environment interactions, which are not fully understood. Neurobiologically, ASD has been associated with abnormalities in synaptic connectivity in the brain, and mutations in the neurexin cell adhesion protein family have been identified in individuals with ASD. The *Nrxn1* +/- mouse is a novel knock-out model of ASD that carries only one functional allele of neurexin 1. The behavioural phenotype of this mouse is unknown and knowledge of the behavioural phenotype is essential because the diagnosis of ASD is based on behavioural symptoms. In this study we tested 20 *Nrxn1* +/- mice (10M, 10F) and 20 C57BL/6J wild-type control mice (10M, 10F) at two months of age in a behavioural test battery. This included 16 tasks that evaluated cognitive and motor performance, mood and anxiety, stereotypical behaviour, social interaction, and visual ability. While the *Nrxn1* +/- mice showed no cognitive or social deficits compared to the wild-type mice, they performed worse on some motor tests and showed increased levels of anxiety. These preliminary results give some indication of the behavioural effects of the expression of neurexin 1. We are currently extending our test battery and testing mice at other ages to determine if they show age-related changes in their behavioural phenotype.

TITLE: Why Don't We Spend Time In Nature?: A Thematic Theory-Driven Analysis

AUTHORS: Sara Cochrane and Shannon Johnson

ABSTRACT: Spending time in nature has been shown to have several health-enhancing benefits, including improved physical health, mental health, cognitive functions, and overall well-being. However, despite a stated interest in spending time outside and in natural environments, people spend only 5% of their time in nature. The objective of this study was to better understand why people do not spend time in nature. We developed a semi-structured interview based on the Theory of Planned Behaviour (TPB) to examine beliefs and attitudes, as well as subjective norms, intentions, and perceived behavioural control about health-related behaviours, and specifically, spending time in nature in an undergraduate sample. Thematic analysis was employed to examine the interview responses. An improved understanding of nature disconnection will aid in the development of behaviour change interventions that will target increasing time spent in nature in order to improve mental, physical, and cognitive health.

TITLE: The Emergence of Envy and *Schadenfreude* Through Social Comparison and the Disappointing Gift Paradigm

AUTHORS: Brianne Derby, Amanda Hudson, & Sophie Jacques

ABSTRACT: Previous studies have found that social emotions such as *envy* and *Schadenfreude* typically emerge in situations of social comparison. The purpose of our study was to examine children's ability to control their displays of these negative social emotions. While envy is considered to be discontent with other's successes, *Schadenfreude* is the pleasure of another's misfortune. We hypothesized that older children would be better at masking social emotions. Younger children aged 49 to 78 months and older children aged 78 to 132 months were tested using the disappointing gift paradigm in situations of social comparison. Results found that older children were more advanced at masking emotions of envy and Schadenfreude whereas younger children were more influenced by social comparison in the disappointing gift paradigm. Future research should compare children's reactions in the disappointing gift paradigm when comparing themselves to adults in contrast to other children.

TITLE: Seeing the forest for the trees in large cognitive datasets

AUTHORS: Richard S. Drake, Sean P. MacKinnon, & Gail A. Eskes

ABSTRACT: Computerized cognitive assessments produce a rich set of data that can be difficult to condense. Each variable typically generates two scores, response speed and accuracy, and many variables are needed to adequately measure performance on target constructs (e.g., subcomponents of working memory or attention). Experimental manipulations such as variable response-stimulus intervals further expand the dataset. Using the output of the Dalhousie Computerized Attention Battery (DalCAB) as an exemplar, we showcase a method of maintaining and manipulating a database, selecting and combining relevant variables, using planned contrasts to visualize the overall pattern of results, and testing for differences using a multilevel modelling approach wherein individuals are nested within variables. The described approach may be used in both single-measurement and intervention (pre-post) research. Using our method, we report the results of a three-week intervention study comparing a dual n-back working memory training game to passive and active control groups.

TITLE: Relating Pupillometric and Behavioural Indices of Effort in Item-Method Directed Forgetting

AUTHORS: Evan F. Forward, Julia C. M. Greenham, Marcus V. Alves, Tracy L. Taylor, Jonathan M. Fawcett

ABSTRACT: Forgetting in an item-method directed forgetting paradigm has been shown to slow responses in the period immediately following the forget instruction and to facilitate removal of attention from the representation of the preceding study item, including its spatial location. These findings have been interpreted as evidence of an active control process associated with the cessation of rehearsal. However, recent findings using pupillometry have also produced conflicting evidence favouring relatively reduced processing demands immediately following the forget instruction as evidence by relatively larger pupil dilation following the remember instruction. The current experiment sought to compare these findings directly by integrating both pupillometry and a secondary probe response task into the same directed forgetting task. Both effects replicated: In addition to a significant directed forgetting effect – with greater recognition of remember than forget items at test – we observed longer response times following forget instructions but larger pupil dilations following remember instructions. We discuss ways in which these findings may be integrated into a more coherent theoretical framework.

TITLE: Searching for a Target: Do Older Adults Check the Template More Often?

AUTHORS: Samantha Good, Jasmine Aziz, Gail Eskes

ABSTRACT: Some cognitive processes important for daily tasks deteriorate with age, including working memory and visual search abilities. Working memory is required to maintain a target template (i.e., image of target object) in mind during visual search. We examined age differences in visual reliance on a target template displayed during search to enhance our knowledge of changes in visual search ability with age. Ten younger (18-35 yrs) and 38 older (55-80 yrs) participants (to date) completed a visual search task while eye movements were monitored. We manipulated the number of target features (one or three) and the number of distractors in the visual scene. Older adults were expected to show increased reliance on the target template (i.e., more eye movements towards the template and longer fixations) than younger adults, particularly as the number of features and distractors increased. Results will provide further knowledge on the changes in cognitive functions with age and can be implemented to promote maintenance of cognitive performance across the life-span.

TITLE: Language and cognition in dogs (*Canis familiaris*)

AUTHORS: Linda Hutchinson, Gabrielle Moulton, Oliver Schnare

ABSTRACT: In human infants, a correlation has been found between the development of language and object permanence, or the ability to recognize that objects still exist when they cannot be seen. Like human children, dogs present ample variability in both language ability and level of object permanence, making them an optimal comparison species for determining whether the correlation

between two variables extends to another species. For this study, language ability was measured through the number of words a dogs recognizes, as reported by their owners in a survey. Cognition was measured through a search game comprised of tasks of increasing complexity that challenged dogs' levels of object permanence. We also examined links between dog breed and attachment on object permanence. We report findings related to these research questions and propose future directions.

TITLE: The Influence of Varying Foreperiods on Error Rate and Reaction Time

AUTHORS: Aydan Idrisova, Kiyana Kamali, Olivia Macdonald, Briana G. Ross, Brett B.T. Feltmate, Colin R. McCormick & Ralph S. Redden

ABSTRACT: Attention is defined as the process of focusing on one stimulus while simultaneously ignoring others. In 1973, Posner, Summers, Klein, and Buggie investigated how a warning signal affects alertness and accuracy. We aimed to replicate their first experiment which provided evidence of speed-accuracy tradeoff and confirmed that rapid response times did not improve accuracy. Reaction time and error rate were measured during the manipulation of foreperiods, as well as compatible or incompatible response conditions. Seventeen subjects participated. The response condition was cued by the border of the screen. A warning signal was played prior to the presentation of the target stimulus. The subjects responded to the stimulus by pressing the key on the same side as the target, or the opposite side. Error rate was the smallest for shorter foreperiods but these reaction times were the longest, therefore supporting the speed-accuracy tradeoff found by Posner et al.

TITLE: Even Further Evidence in Favor of Prior Entry from Endogenous Attention to a Location in Space: A Replication Study

AUTHORS: Kirsten M. Lebel, Morgan G. MacIntosh, Samantha S. Tse, Sana M. Fashir, Brett B.T. Feltmate, Colin R. McCormick & Ralph S. Redden

ABSTRACT: The laws of prior entry state that attended objects are perceived faster than unattended objects. Previous research has provided contradictory results about the presence of prior entry (Shore et al., 2001; Schneider & Bavelier, 2003). The goal of our current experiment was to replicate a previous study by Redden et al. (2017) which incorporated an orthogonal color probe task into the original TOJ paradigm in order to settle the discrepancy in previous literature and determine if prior entry exists under endogenous cueing. We hypothesized that prior entry would show effects under the endogenous cueing condition. The present study used a modified TOJ task with an orthogonal color probe task to further reduce any response biases and assess prior entry independent of response mechanisms. Effects of attention on prior entry and probability of color encoding were replicated from Redden et al. (2017). Other findings from the experiment will be discussed.

TITLE: On the independence of attentional mechanisms in temporal orienting: A replication study

AUTHORS: Kate MacDiarmid, Fatema Al-Shammary, Cathryn LeBel, Isabella Rossi

ABSTRACT: Attention has been theorized to be composed of two networks: the exogenous network, which involves bottom-up processing; and the endogenous network, which involves top-down processing. A replication of a 2018 study by McCormick and colleagues was performed to provide further evidence for the independence of endogenous and exogenous networks in temporal attention (attention in time). We hypothesized that endogenous and exogenous temporal attention would be found to be independent from one another, as demonstrated by the effects of two measures: cue type and signal type. Fifteen participants performed a computerized task that measured reaction time as these two measures were manipulated. Data analysis revealed a significant effect of both cue and signal independence, as well as the cue and signal variance. Implications and limitations are discussed.

TITLE: Reading comprehension, prosodic sensitivity, and the role of prosodic cues in text

AUTHORS: Alexandra M. Ryken, S. Hélène Deacon, and Lesly Wade-Woolley

ABSTRACT: Reading comprehension, or the ability to understand the meaning of text, is one of the most important skills children learn in school. Understanding how various language skills support children's reading comprehension is essential to supporting early literacy development. One language skill which is related to reading comprehension in young readers is prosodic sensitivity, which is an awareness of the rhythmic cues in speech. However, the reason for this relationship is unclear. This study tested the possibility that children's awareness of how rhythmic cues are represented in text, e.g. through punctuation, may mediate the relationship between prosodic sensitivity and reading comprehension. Approximately 150 students in grades 3-5 were assessed, and awareness of prosodic cues in text did emerge as a partial mediator in the relationship between prosodic sensitivity and reading comprehension. Children may be using their awareness of rhythmic cues in text when attempting to understand what they read.

TITLE: Knowledge of Spoken Words by Domestic Dogs: A New Instrument for Use With Dog Owners

AUTHORS: Catherine Reeve & Sophie Jacques

ABSTRACT: Dogs are sensitive to human social cues (Miklosi & Topal, 2013) and can interpret physical and verbal cues from humans to gain information about new situations (Merola et al., 2012), or successfully complete tasks (Kaminski & Nitzschner, 2013). No reliable tool exists for determining which and how many words dogs know. The current study included a new measure designed to assess the number of words to which dogs respond differentially and systematically. Our tool was modeled after the MacArthur-Bates CDI (Fenson et al., 1994) for use with preverbal infants. Our first goal was to assess the content validity and usability of our online instrument, while our second was to use owners' responses to identify predictors of dogs' word understanding. Participants included

168 owners of dogs of various ages and breeds. Owners answered questions about themselves and their dog, and identified and rated the level at which their dog responded to each word in a checklist. Owners reported that their dogs responded selectively to 18 to 202 ($M=89$ words; $SD=36$) and variables such as dog owners' experience with dog training, how quickly dogs reportedly learned new tricks, and breed type showed predictive utility. Principal component analyses revealed that words to which dogs respond loaded onto 3 distinct, yet related components: *general words*, *verbs*, and *nouns*. Our tool not only allows for a standardized measurement of dogs' receptive abilities, but may eventually also become an important tool used to predict the potential of working dogs.

TITLE: Forgetting on purpose; How and what do we intentionally forget?

AUTHORS: Kaelen Watters, Anjali Pandey, Tracy Taylor-Helmick

ABSTRACT: This study examined the extent to which the visual characteristics of to-be-remembered and to-be-forgotten words are encoded into long-term memory. Using an item-method directed forgetting task, participants completed a study phase consisting of list of words presented one at a time, each followed by an instruction to remember or forget. In a subsequent test phase, participants then made yes-no recognition decisions to: 1) 'remember' and 'forget' words from the study phase; 2) unstudied foil words; and, 3) unstudied foil words that looked similar to the 'remember' and 'forget' words. We expected a directed forgetting effect, with greater recognition of 'remember' words than 'forget' words. However, our main interest was in determining whether visual similarity to *studied* words increases the false alarms made to *unstudied* words and, if so, whether this tendency is influenced by whether the unstudied word looks similar to a 'remember' item or a 'forget' item.

TITLE: Investigating the neural bases of self-monitoring of speech in people with aphasia: an ERP study

AUTHORS: Francesco Usai, Heather Ouellette and Aaron J. Newman

ABSTRACT: Aphasia is a communication disorder that can impair the ability to find the right words, name things correctly, and to monitor one's own speech for the presence of errors. Little is known about the neural bases of self-monitoring of speech in aphasia. To address this gap, we recorded EEG from people with aphasia (current $n=2$) while performing a picture-naming task, and observed the effect of response accuracy on the error-related negativity (ERN), an EEG effect associated with performance monitoring across different cognitive domains. The time course and scalp topographies of EEG were contrasted for incorrect and correct responses. For one participant, this contrast resembled the ERN, while the other showed an effect with a similar time course but with opposite polarity (i.e., a positivity). These differences may be explained by the different neuropsychological profile of the two subjects, although these preliminary results provide encouraging evidence that the ERN may be a useful marker of self-monitoring in aphasia.